

6,906: Economics: Macro for Computer Science

LECTURE 7: SHORT-RUN FLUCTUATIONS AND THE IS-LM MODEL

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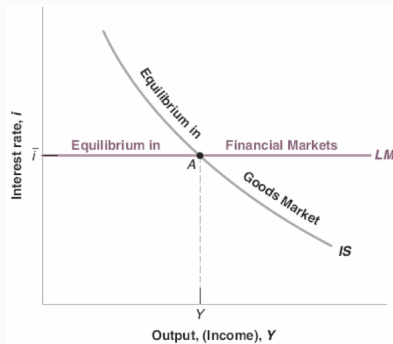
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Learning objectives

- By the end of this lecture, you should be able to:
 - Use one of the workhorse models of macroeconomics to analyze the short run;
 - Discuss policy options in the basic and the extended IS-LM models.

IS-LM model

- The IS relation:
$$Y = C(Y - T) + I(Y, i) + G.$$
- The LM relation as an interest rate rule:
$$i = \bar{i}.$$
- The IS and LM relations together determine output.
- Any point on the IS curve corresponds to equilibrium in the goods and services. Any point on the LM curve corresponds to equilibrium in financial markets. Only at their intersection (point A) are both equilibrium relations satisfied.



Source: Blanchard (2021)

Policy options in the IS-LM model

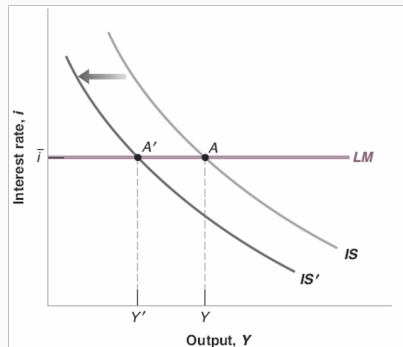
- Some terms:
 - $G - T < 0 \Leftrightarrow$ budget surplus
 - $G - T > 0 \Leftrightarrow$ budget deficit
 - $G - T = 0 \Leftrightarrow$ balanced budget
- Fiscal policy
 - Tax reduction/increase: e.g., income tax, value added tax, capital gains tax, corporate tax, etc.
 - Transfer payments reduction/increase: e.g., unemployment benefits, etc.
 - Government spending reduction/increase: e.g., research and education, military equipment, infrastructure, etc.

Policy options in the IS-LM model

- Fiscal policy
 - Decrease in $G - T \Leftrightarrow$ fiscal contraction $\Leftrightarrow$ fiscal consolidation
 - Increase in $G - T \Leftrightarrow$ fiscal expansion
- Monetary policy
 - Decrease in $i \Leftrightarrow$ increase in $M \Leftrightarrow$ monetary expansion
 - Increase in $i \Leftrightarrow$ decrease in $M \Leftrightarrow$ monetary contraction $\Leftrightarrow$ monetary tightening
- Steps for analyzing the effects of changes in policy or exogenous variables:
 - Does it shift the IS curve and/or the LM curve?
 - What does this do to equilibrium output and the equilibrium interest rate?
 - Describe the effects in words.

Fiscal contraction in the IS-LM model

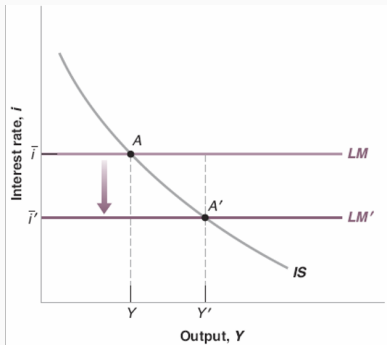
- An increase in taxes shifts the IS curve to the left.
- This leads to a decrease in the equilibrium level of output.
- Transmission mechanism:
 - $T \nearrow \Rightarrow Y_D \searrow \Rightarrow C \searrow \Rightarrow$ demand for goods $\searrow \Rightarrow$ production $\searrow \Rightarrow Y \searrow \Rightarrow$ multiplier effect
 $\Rightarrow Y \searrow \Rightarrow M^d \searrow \Rightarrow M^s \searrow \Rightarrow i$ unchanged
- Should the government better increase taxes or reduce spending?



Source: Blanchard (2021)

Monetary expansion in the IS-LM model

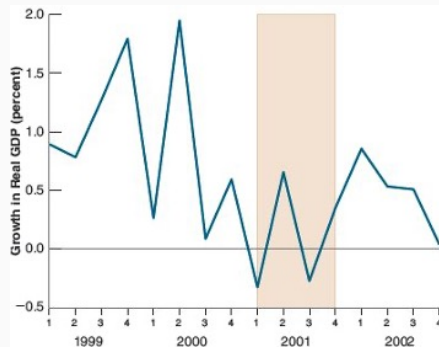
- A decrease in the interest rate shifts the LM curve down.
- This leads to higher equilibrium output.
- Transmission mechanism:
 - $M^s \nearrow \Rightarrow i \searrow \Rightarrow I(i, Y) \nearrow \Rightarrow$ demand for goods $\nearrow \Rightarrow Y \nearrow \Rightarrow$ multiplier effect $\Rightarrow Y \nearrow \Rightarrow M^d \nearrow \Rightarrow M^s \nearrow$



Source: Blanchard (2021)

Case study: Policy response to the U.S. recession of 2001

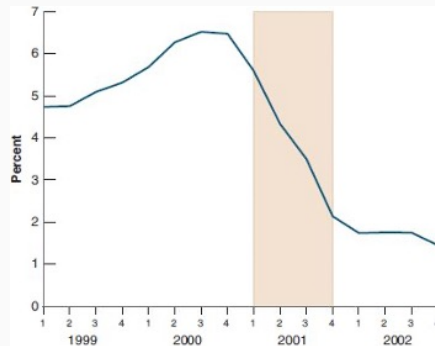
- The National Bureau of Economic Research concluded that the U.S. economy was in a recession between March 2001 and December 2001, triggered by sharp declines in investment demand.



Source: Blanchard (2021)

Case study: Policy response to the U.S. recession of 2001

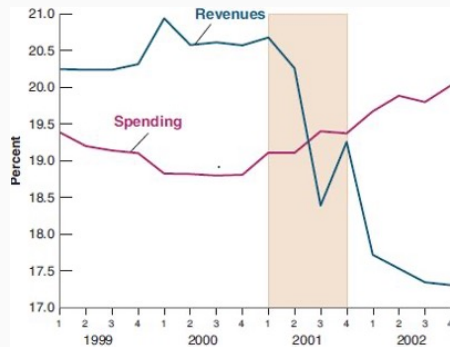
- The Fed cut the federal funds rate from 6.5% in January to 2% at the end of 2001.



Source: Blanchard (2021)

Case study: Policy response to the U.S. recession of 2001

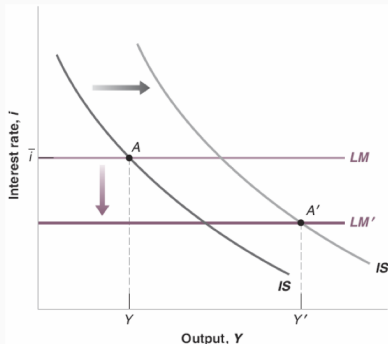
- They also introduced tax cuts in the 2001 and 2002 budgets. September 11, 2001, also led to an increase in defense and homeland security spending.



Source: Blanchard (2021)

Monetary-fiscal policy mix in the IS-LM model

- Monetary-fiscal policy mix is the combination of monetary and fiscal policies.
- When the economy is in a recession and output is too low, both fiscal and monetary policies can be used to increase output.
- The fiscal expansion shifts the IS curve to the right.
- A monetary expansion shifts the LM curve down.
- Both lead to higher output.



Source: Blanchard (2021)

- The effects of fiscal and monetary policy may be weaker than predicted by the IS-LM model.
 - Criticism of a fiscal policy expansion - time lag (effect may be delayed), people expect higher taxes in the future, multiplier reduced by savings and imports, etc.
 - Criticism of monetary policy expansion - generate inflation, liquidity trap, credibility of the central bank, effect on exchange rates etc.
- As with any economic model, the IS-LM has limitations, but it can be extended. Possible extensions:
 - Introduce risk premia
 - Introduce expectations (e.g. consumption and investment depend on expectations)
 - Introduce open economy elements: exports, imports, exchange rates (not covered in this course)

The extended IS-LM model

- Until now, we assumed that there were only two financial assets-money and bonds-and just one interest rate-the rate on bonds-determined by monetary policy.
- The financial system also plays a major role in the economy.
- Nominal interest rate is the interest rate in terms of money (eg. dollars).
- Real interest rate is the interest rate in terms of a basket of goods.
- We must adjust the nominal interest rate to take into account expected inflation.

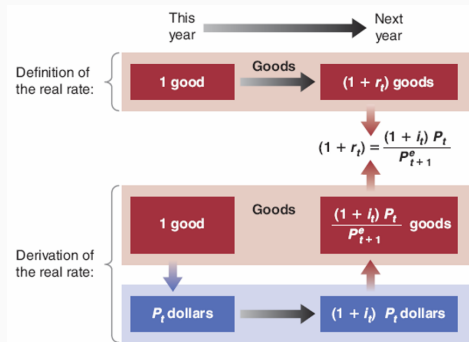
The extended IS-LM model

- If the nominal interest rate, r_t , and expected inflation π_{t+1}^e are not too large, the relationship can be expressed by:

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e} \approx r_t = i_t - \pi_{t+1}^e.$$

- When expected inflation equals zero, the nominal interest rate and the real interest rate are equal.
- Because expected inflation is typically positive, the real interest rate is typically lower than the nominal interest rate.
- For a given nominal interest rate, the higher the expected inflation, the lower the real interest rate.

Nominal versus real interest rate



Source: Blanchard (2021)

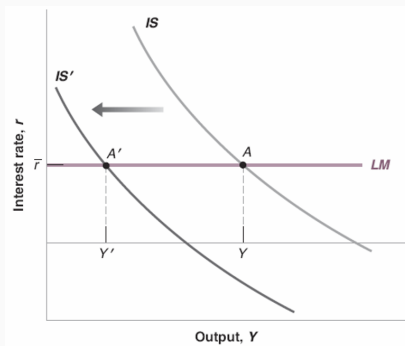
- Some bonds are risky, so bondholders require a risk premium.
- Let i be the nominal interest rate on a riskless bond, x be the risk premium, and p is the probability of default, then to get the same expected return on the risky bonds as on the riskless bond:

$$(1 + i) = (1 - p) \cdot (1 + i + x) + (p) \cdot (0)$$

- so that $x = (1 + i) \cdot p / (1 - p)$.
- Risk premia are determined by:
 - The probability of default
 - The degree of risk aversion of bondholders

Extended IS-LM model

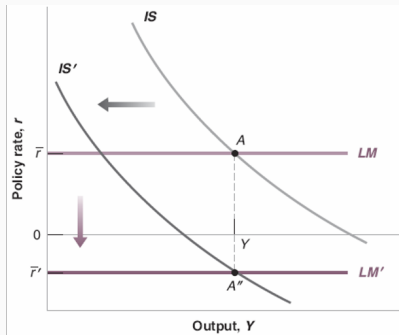
- The central bank now chooses the real policy rate r , which enters the IS equation as part of the borrowing rate $(r + x)$ for consumers and firms:
- The IS relation:
$$Y = C(Y - T) + I(Y, r + x) + G.$$
- The LM relation as an interest rate rule:
$$r = \bar{r}.$$
- A financial shock, meaning an increase in x , leads to a shift of the IS curve to the left and a decrease in equilibrium output.



Source: Blanchard (2021)

Extended IS-LM model

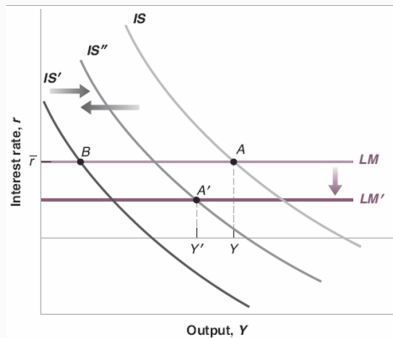
- If sufficiently large, a decrease in the policy rate can in principle offset the increase in the risk premium.
- The zero lower bound may limit the decrease in the real policy rate.



Source: Blanchard (2021)

The Financial Crisis, and the use of financial, fiscal, and monetary policies

- The financial crisis led to a shift of the IS to the left.
- Financial and fiscal policies led to some shift back to the IS to the right.
- Monetary policy led to a shift of the LM curve down.
- Policies were not enough however to avoid a major recession.



Source: Blanchard (2021)

- Today we learned:
 - To use the IS-LM model to study economic policy for the short run.
- However, in the medium run resource constraints are important and inflation may occur, e.g. due to expansionary monetary policy. Therefore, we examine the labor market and inflation in the next class.